

Which Songs Overperform or Underperform on Spotify?

Project idea and machine-learning pipeline

Core question: Which songs perform much better or much worse on Spotify than we would expect from their popularity elsewhere?

1. Project idea

The 2024 Spotify dataset contains Spotify streaming performance together with measures of song popularity and exposure from other sources, such as YouTube, TikTok, Shazam, radio/AirPlay, Pandora, and playlist metrics. Instead of asking only whether these variables can predict Spotify streams, this project uses prediction as a tool to identify **unusual songs**.

The basic idea is to estimate how many Spotify streams a song would be expected to have, given its non-Spotify popularity profile. Then we compare that prediction with the song's actual Spotify streams.

Song	Predicted Spotify streams	Actual Spotify streams	Interpretation
A	300M	800M	Strong overperformer
B	600M	200M	Strong underperformer
C	450M	470M	Close to expected

The key statistical quantity: the residual

For song i , define the residual as:

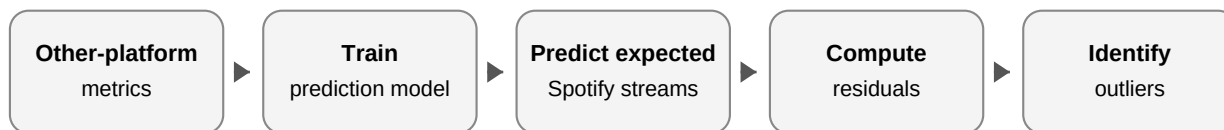
$$e_i = \text{Actual Spotify Streams}_i - \text{Predicted Spotify Streams}_i.$$

A large positive residual means the song is performing much better on Spotify than the model would expect. A large negative residual means the song is performing much worse on Spotify than expected, given its other engagement measures.

Why this is more interesting than ordinary prediction

A standard prediction project would stop after reporting which model gives the best RMSE or R-squared. Here, prediction is only the first step. The real research question is about the **exceptions**: which songs behave differently from the general pattern, and what characteristics distinguish them?

2. Machine-learning pipeline



Step	What happens	Purpose
1. Choose outcome	Use Spotify streams as the response variable; log-transform if strongly skewed.	Creates the quantity to be predicted.
2. Choose predictors	Use non-Spotify metrics such as YouTube, TikTok, Shazam, AirPlay/radio, and Pandora.	Avoids using Spotify-derived measures to predict Spotify itself.
3. Clean + transform	Handle missing values and log-transform large count variables where appropriate.	Reduces skew and improves numerical behavior.
4. Split data	Create training/test sets or use cross-validation.	Measures out-of-sample predictive performance.
5. Fit baseline	Fit a multiple linear model or Elastic Net.	Provides an interpretable benchmark.
6. Fit ML models	Fit Random Forest and/or XGBoost.	Captures nonlinearities and interactions.
7. Compare models	Compare RMSE, MAE, and R-squared.	Selects the model that best estimates expected Spotify performance.
8. Predict expected streams	Generate predicted Spotify streams for each song.	Creates the benchmark level for each song.
9. Compute residuals	Subtract predicted streams from actual streams.	Measures overperformance or underperformance.
10. Investigate outliers	Rank the largest positive and negative residuals.	Identifies the most unusual songs for interpretation.

Candidate models

Elastic Net: useful when many predictors are correlated and we want a regularized, interpretable baseline.

Random Forest: captures nonlinear relationships and interactions without specifying them in advance.

XGBoost: often performs well on structured tabular data and can capture complex patterns in engagement metrics.

Important modeling choice

If the goal is to ask whether a song is unusually strong on Spotify relative to its popularity elsewhere, the predictors should avoid variables that already encode Spotify performance, such as Spotify popularity, Spotify playlist reach, or all-time rank if those quantities are partly determined by Spotify success. Using them could create data leakage and make the residuals less meaningful.

3. What the final analysis could show

The clearest visualization would be a plot of **predicted Spotify streams** against **actual Spotify streams**. Songs close to the diagonal line behave approximately as expected. Songs far above or below the line are the interesting cases.

Possible result	Interpretation
Large positive residual	Spotify overperformer: actual streams are much higher than expected.
Large negative residual	Spotify underperformer: actual streams are much lower than expected.
Residual near zero	Spotify performance is broadly consistent with the song's other engagement metrics.

Possible follow-up questions

After identifying the largest outliers, the project could investigate whether overperformers and underperformers differ systematically by release year, artist, explicit status, platform-specific engagement, or other available characteristics. These would be secondary analyses, not assumptions built into the model.

Concise group pitch

"My backup idea is to identify songs that overperform or underperform on Spotify relative to their popularity elsewhere. We could use non-Spotify engagement variables such as YouTube, TikTok, Shazam, radio/AirPlay, and Pandora to predict expected Spotify streams using models such as Elastic Net, Random Forest, or XGBoost. Then we would calculate residuals - actual minus predicted streams - and use those residuals to find the songs whose Spotify performance is unusually high or low. The main goal would not just be prediction, but understanding the biggest exceptions to the general pattern."

Scope note

This is an observational project. A large residual does not prove that any external platform caused Spotify success or failure. It only identifies songs whose Spotify performance differs substantially from what would be expected based on the selected predictors.

Repository: <https://github.com/megmillet/Spotify-Project>